Test II

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Novel Prep

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Phase 3 · Test II

Overview

Test II

24 word-problem training challenges

Question 1

For the equation, b and c are constants. For any real number t, the point $\left(\dfrac{-2t+4}{3}, t\right)$ lies on the graph of the equation $bx + 3y = c$. What is the value of c?

A. -2

B. 4

C. 6

D. 12

Answer 1

Correct Answer: C

Question 2

The function q is defined by

$$q(x) = a|x-9|^2 - 87|x-9| + b,$$

where a and b are constants and $a > b > 87$. In the xy-plane, the graph of $y = q(x)$ contains the points $(639, h)$ and $(-621, k)$, where h and k are constants. What is the value of $9(-639)^{h-k} + 621(9)^{k-h}$?

Answer 2

Final Answer: $\boxed{630}$

Question 3

In the given equation, a and b are positive constants. A solution to the equation is $x = 42$.

$$x(9x - 2a) + b(9x - 2a) - (x + b) = 0$$

What is the value of a?

Answer 3

Final Answer: $\boxed{188.5}$

Question 4

The function z is defined by

$$z(w) = (0.829)^{2w}.$$

The value of $z(w)$ decreases by $p\%$ for each increase by 1 in the value of w. Which of the following is closest to the value of p?

A. 0.171

B. 0.829

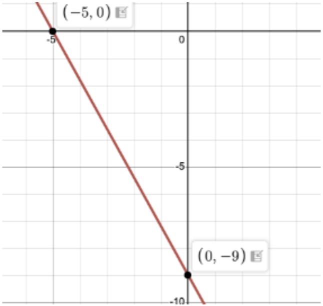
C. 17.1

D. 31.3

Answer 4

Correct Answer: D

Question 5



The graph of line h is shown in the xy-plane. Line k (not shown) is defined by the equation $sx + 40y = t$, where s and t are constants. If line k is graphed in this xy-plane, the result is a graph of two linear equations. This system of two linear equations has no solution. Which of the following is NOT a possible value of t?

A. -360

B. -9

C. 72

Answer 5

Correct Answer: A

Question 6

The function h is defined by

$$h(x) = -\sqrt{x^2 + bx + c},$$

where b and c are constants. The graph of $y = h(x)$ contains the points $(2, 0)$ and $\left(0, -\sqrt{358}\right)$. If $h(m) = 0$, what is the greatest possible value of m?

A. 2

B. 177

C. 179

D. 358

Answer 6

Correct Answer: C

Question 7

The functions f and g are defined by the equations shown, where a and b are integer constants, $a < b$, and $b < 0$. If $y = f(x)$ and $y = g(x)$ are graphed in the xy-plane, which of the following equations displays, as a constant or coefficient, the y-coordinate of the y-intercept of the graph of the corresponding function?

I. $f(x) = a(2.1)^{x+b}$

II. $g(x) = a(2.1)^{x} + b$

A. I only

B. II only

C. I and II

D. Neither I nor II

Answer 7

Correct Answer: D

Question 8

The function f is defined by

$$f(x) = \frac{x^2 + ax + b}{2x + c},$$

where a, b, and c are constants. The graph of $y = f(x)$ does not intersect the line $x = 4$. If $f(5) = f(6) = 0$, what is the value of $a + b + c$?

A. 10

B. 11

C. 12

D. 13

Answer 8

Correct Answer: B

Question 9

The positive number a is 2,106% of the sum of the positive numbers b and c, and b is 81% of c. What percent of b is a?

A. 4,706%

B. 47.06%

C. 470,600%

D. 47,060%

Answer 9

Correct Answer: A

Question 10

Point N lies on a unit circle in the xy-plane and has coordinates $(1,0)$. Point O is the center of the circle and has coordinates $(0,0)$. Point M also lies on the unit circle, and the measure of angle $\angle NOM$ is $\dfrac{266\pi}{3}$ radians. If the coordinates of point M are (a,b), where a and b are constants, what is the value of a?

A. $\dfrac{1}{2}$

B. $\dfrac{\sqrt{3}}{2}$

C. $-\dfrac{1}{2}$

D. $-\dfrac{\sqrt{3}}{2}$

Answer 10

Correct Answer: C

Question 11

In the given equation, a and b are positive constants. The sum of the solutions to the given equation is $k(6a + b)$, where k is a constant. What is the value of k?

$$24x^2 - (12a + 2b)x + ab = 0$$

A. $\frac{1}{12}$

B. $-\frac{1}{12}$

C. 12

D. -12

Answer 11

Correct Answer: A

Question 12

In triangle ABC, the measure of angle B is 90° and BD is an altitude of the triangle. The length of $AB = 15$, and the length of AC is 23 greater than the length of AB. What is the value of $\dfrac{BC}{BD}$?

A. $\dfrac{15}{38}$

B. $\dfrac{15}{23}$

C. $\dfrac{23}{15}$

D. $\dfrac{38}{15}$

Answer 12

Correct Answer: D

Question 13

A right square pyramid has a height of 11 centimeters (cm). A second right square pyramid has a height of 22 centimeters. The area of the bases of each of the two pyramids is $100\,\text{cm}^2$. Which of the following is closest to the difference in the surface area of the second pyramid and the surface area of the first pyramid, in cm^2?

A. 209

B. 220

C. 367

D. 551

Answer 13

Correct Answer: A

Question 14

A researcher investigated two species of mites: a predator and its prey. At the start of a week, there was an equal number of the two species. At the end of the week:

- The number of prey had increased by 2,700% of the number of prey at the start of the week.
- The number of predators had increased by 180% of the number of predators at the start of the week.

The number of prey at the end of the week was $p\%$ greater than the number of predators at the end of the week. What is the value of p?

A. 9

B. 90

C. 900

D. 9,000

Answer 14

Correct Answer: C

Question 15

A beekeeper's initial observation of the population of a certain bee colony was 1,100 bees. The beekeeper set a goal to increase the population to 2,600 bees. The beekeeper uses a model that predicts the population of this bee colony:

- begins at 1,100
- increases by 120 bees per week in the first two weeks
- then increases by 180 bees per week until the goal is reached

According to this model, at the end of week w after the initial observation, where $w > 2$, which of the following functions gives the predicted number of bees still needed to reach the beekeeper's goal?

A. $p(w) = 2600 - 180w$

B. $p(w) = 2480 + 180w$

C. $p(w) = 1620 - 180w$

D. $p(w) = -120 + 180w$

Answer 15

Correct Answer: C

Question 16

The speed of a vehicle is increasing at a rate of 7.3 meters per second squared. What is this rate, in miles per minute squared, rounded to the nearest tenth? (Use 1 mile $= 1{,}609$ meters.)

A. 0.3

B. 16.3

C. 195.8

D. 220.4

Answer 16

Correct Answer: B

Question 17

Which of the following expressions has a factor of $x + 2b$, where b is a positive integer constant?

A. $3x^2 + 7x + 14b$

B. $3x^2 + 16x + 14b$

C. $3x^2 + 18x + 14b$

D. $3x^2 + 25x + 14b$

Answer 17

Correct Answer: D

Question 18

In the given system of equations, t is a constant. If the system has no solution, what is the value of t?

$$16x - 20y = 6y + 8$$
$$ty = 8x + \frac{1}{5}$$

Answer 18

Final Answer: $\boxed{13}$

Question 19

In the given equation, k is a positive constant. The product of the solutions to the equation is 72. What is the value of k?

$$\frac{5}{9}(5x+9)\left(x+\sqrt{5k+9}\right)\left(x-\sqrt{5k+9}\right)=0$$

Answer 19

Final Answer: $\boxed{6.2}$

Question 20

In the xy-plane, a unit circle with center at the origin O contains point A with coordinates $(1, 0)$ and point B with coordinates $\left(\frac{5}{\sqrt{34}}, -\frac{3}{\sqrt{34}}\right)$. If the measure of angle AOB is p radians, what is the value of $\frac{\cos p}{\sin p}$?

Answer 20

Final Answer: $\boxed{\dfrac{-5}{3}}$

Question 21

A pyramid with square base area of $36\,\mathrm{m}^2$ has a total surface area of $36 + 12\sqrt{493}\,\mathrm{m}^2$. Find the height of the pyramid.

Answer 21

Final Answer: $\boxed{22}$

Question 22

The height of a right circular cylinder is 36 inches, and the circumference of its base is 360 inches. Which expression represents the total surface area, in square inches, of the cylinder?

A. $(36)(360) + 2\pi\left(\frac{180}{\pi}\right)^2$

B. $(36)(360) + \pi\left(\frac{180}{\pi}\right)^2$

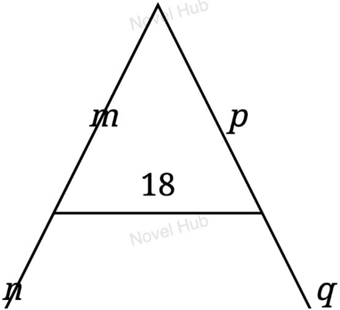
C. $\pi\left(\frac{180}{\pi}\right)^2(36)$

D. $(36)(360)$

Answer 22

Correct Answer: A

Question 23



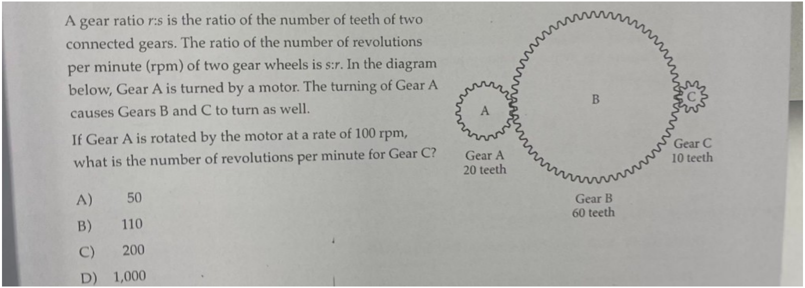
The parallel line segment divides the other two sides of the triangle into segments of lengths m and n, and p and q, respectively. If $\frac{p}{q} = k$ and $\frac{m}{n} = k$, what is the value of k?

Answer 23

Final Answer: $\boxed{1.5}$

Question 24

A gear ratio $r : s$ is the ratio of the number of teeth of two connected gears. The ratio of the number of revolutions per minute (rpm) of two gear wheels is $s : r$. In the diagram below, Gear A is turned by a motor. The turning of Gear A causes Gears B and C to turn as well.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If Gear A is rotated by the motor at a rate of 100 rpm, what is the number of revolutions per minute for Gear C?

A) 50

B) 110

C) 200

D) 1,000

If Gear A is rotated by the motor at a rate of 100 rpm, what is the number of revolutions per minute for Gear C?

A. 50

B. 110

Answer 24

Correct Answer: C

Thank You!

We appreciate your time and focus.

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